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A novel predictive-reactive rescheduling method for products assembly lines with optimal dynamic pegging

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ABSTRACT

In this paper a new predictive-reactive rescheduling method is presented based on the dynamic pegging concept. Scheduling adjustments in a multi-level, mixed-model production system are invoked in response to the unplanned events, such as new order arrival, changes in due dates, and order cancellation. By using the dynamic pegging approach at the time of rescheduling, Work-In-Progress (WIP) can be reassigned to the existing or new orders. Based on our previous work on reactive-only rescheduling with dynamic pegging, a Mixed Integer Programming (MIP) model is developed for predictive-reactive scheduling of new orders of different product types that arrive following exponential distribution. A simple example is designed to compare the performance of predictive-reactive rescheduling and the reactive-only rescheduling by changing different input parameters to the MIP model. The results indicate that in certain scenarios implementing the predictive-reactive approach could be significantly more cost-efficient.

1. Introduction

Integrating production planning and scheduling has always been an important topic of interest. Material Requirements Planning (MRP) and Manufacturing Resource Planning (MRPII) are some of the most popular Manufacturing IT solutions that have been implemented for efficient resource management in different companies. These particular software however had a main drawback of not considering capacity constraint and therefore creating infeasible production schedules (Chen and Ji, 2007b). In this regard, more advanced IT solutions such as Advanced Planning and Scheduling (APS) were proposed which took the capacity constraint and sequence of operations into account (Chen and Ji, 2007b, Steger-Jensen et al., 2011).

While these solutions had significant advantages over the previous ones, their generated production schedules were often infeasible when disruptions such as new order arrival, due dates change, order cancellation, or machine failures occurred in the system (Ouelhadj and Petrovic, 2009). In these situations, adjusting the original schedule through dynamic scheduling and re-scheduling techniques was proven to be an effective mechanism for adaptation (Ouelhadj and Petrovic, 2009, Renke et al., 2021).

By using the dynamic scheduling methods, scholars have been trying to decrease the negative effects of disruptions in manufacturing systems and supply chains. dynamic scheduling aims at maximizing productivity and minimizing deviation from the original schedule *after* either job or resource-related disruptions happen in the system. Occurrence of a new event triggers the dynamic scheduling methods to adjust the original schedule by means of different schedule repair or rescheduling techniques. These adjustments are usually made with respect to a certain objective function; for example, minimizing make span, shop flow time, lateness and/or maximizing system's stability (Ouelhadj and Petrovic, 2009). Some of the most commonly used re-scheduling techniques include completely reactive, robust-proactive, and predictive-reactive (Jiang et al., 2016, Hamzadayi and Yildiz, 2016, Chen and Ji, 2007a, Wu et al., 1993).

Most re-scheduling techniques mainly focus on re-allocating the available resources but other useful methods can also be employed in certain production environments; For example, in manufacturing systems with multi-level mixed-model production, resource shortages caused by a certain disruption can be managed by a technique called delayed product differentiation (Lee and Tang, 1997, Gupta and Benjaafar, 2004, Jafari et al., 2022); a concept in which certain WIP assigned to an order (also known as *pegging*) that is common to all product models are delayed as much as possible to mitigate the effect of

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random disruptions and increase the chances of addressing the final demand.

In most APS systems, pegging is fixed, i.e., it is generated at the beginning of a planning cycle and remains unchanged until a new planning cycle starts. However, some advanced commercial APS systems such as SAP's Advanced Planning and Optimization (APO) implement the capability of *dynamic pegging* (Kallrath and Maindl, 2006). This capability enables changing the job-order assignment if a certain disruption occurs in the system. Like delayed differentiation, dynamic pegging can be quite advantageous in multi-level production since it would create the opportunity of using the work in process (WIP) of an existing job for another job that is more urgent.

As discussed in (Moghaddam and Saitou, 2019, Moghaddam and Saitou, 2020), coupling re-scheduling techniques with dynamic pegging strategy in multi-level mixed-model production systems could be quite beneficial and more effective than using mere re-scheduling techniques. In (Moghaddam and Saitou, 2019), all WIP produced prior to occurrence of a disruptive event can be re-assigned to new orders as long as certain criteria are met and certain constraints are satisfied. It is also assumed that all actions are *reactive* and therefore, re-pegging and re-scheduling are done after a new order arrives unexpectedly. In other words, there is no prior knowledge about the nature of new orders.

Nowadays the APS problem cannot be solved with the assumption of accurate, complete knowledge about the production environment and most importantly order information. With the growing demand for product customization and the shift towards smart manufacturing, planning and scheduling must be performed dynamically and by considering many random and stochastic events in advance. Since data availability has been increased in the past few years, many companies are now able to more accurately predict critical production parameters such as periodical arrivals of new orders.

This paper aims to investigate the effect of data availability and predictions on the previously formulated APS problem by (Moghaddam and Saitou, 2019, Moghaddam and Saitou, 2020) while considering multiple production periods and multiple product types by implementing a predictive-reactive approach. The presented rescheduling/repegging problem contributes to the operations research area by:

- Studying the rescheduling problem combined with dynamic reassignment of WIP in a multi-level mixed-model production system.
- Implementing a predictive-reactive MILP model to match the simultaneous rescheduling/repegging problem.
- Carefully designing simple examples to analyze the performance of the new predictive-reactive formulation in comparison with the previous reactive-only model, while handling different challenging scenarios.
- Providing managerial insights to handle realistic larger scale production problems based on the proposed examples and the comprehensive analysis of the results.

The rest of this paper is organized as follows: In Section 2 a literature review is presented on integrated production planning and scheduling as well as predictive reactive rescheduling. The optimization model, all model assumptions and the proposed methodology are introduced in Section 3. Section 4 presents the comparison of the results under multiple possible scenarios. In Section 5 the results are thoroughly discussed and some managerial insights are provided. Finally, Section 6 entails the conclusion and the areas for future research.

2. Literature review

Undoubtedly, production planning and production scheduling are two closely interconnected problems; in fact, the decisions made on the planning level, in terms of production goals, are usually the inputs for the scheduling level (Georgiadis et al., 2021). Production planning involves the whole manufacturing process on a high level in order to plan the production of finished goods. The main objective in production planning is fulfilling marketing demands at minimum production and inventory costs. In comparison, production scheduling is a more detail-oriented process with a shorter time horizon. The goal is to schedule every single operation at a certain time in the calendar, assign the resources and workers, and plan all the detailed steps in order to optimize certain efficiency criteria.

Each of these processes, separately used, are irreplaceable parts of a successful manufacturing system. However, it has been shown that the integrated and coordinated production planning and scheduling across various levels of a manufacturing supply chain can yield a significantly improved overall economic performance (Castro et al., 2018).

2.1. Integrated production planning and scheduling

The problem of integrated production planning and scheduling, mostly known as the APS problem, is proved to be NP-hard (Lasserre, 1992, Sukkerd and Wuttipornpun, 2016). As a result, simplifying assumptions and approximations were made in many of the previous studies that proposed heuristic (Hastings et al., 1982, Bahl and Ritzman, 1984, Taal and Wortmann, 1997), or mathematical programming approaches (Lasserre, 1992, Dillenberger et al., 1994, Kolisch, 2000, Moon et al., 2004) to solve it.

A summary of the literature on APS problems can be seen in Table 1. As shown in Table 1, in studies of APS, minimizing make-span, cost of idle time, and earliness-tardiness penalties are the most common objective functions (Chen and Ji, 2007b, Örnek et al., 2010, Peng et al., 2014, Ozturk and Ornek, 2016, Moghaddam and Saitou, 2019, Vieira et al., 2021). Many of the previous works proposed mixed integer programming (MIP) formulations for solving the APS problem (Fachini et al., 2018, Chansombat et al., 2019, Georgiadis et al., 2021, Vieira et al., 2021). For example, Chen and Ji (2007b) successfully integrated production planning and scheduling problems in the context of APS by developing a MIP model. Their developed model could be used in production environments where multiple orders are placed for products with multi-level structures. Due dates, capacity constraints, release time, and operation sequences were taken into account in order to minimize the total cost of idle time and tardiness/earliness penalties. This model was later extended by Örnek et al. (2010) to allow operation assignment to multiple machine candidates in the system.

Many of the exact solving approaches became intractable in case of large-scale real world problems and therefore, heuristics, metaheuristics and other integrated frameworks were developed and presented to handle large scale APS problems in different fields of industry such as capital goods, glass container, and automotive parts manufacturing (Sukkerd and Wuttipornpun, 2016, Chen et al., 2017, Chansombat et al., 2019, Vieira et al., 2021).

Integrated production planning and scheduling can prevent the creation of infeasible production schedules for the generated production plans since capacity constraint and sequence of operations are taken into account in the planning phase. However, the dynamic situation and various uncertainties that naturally appear in any manufacturing environment (new order arrival, due dates change, order cancellation, or machine failures) call for adjusting the original schedules through dynamic scheduling and re-scheduling techniques.

2.2. Predictive reactive rescheduling

According to Ouelhadj and Petrovic (2009), dynamic scheduling can be categorized based on three basic strategies; completely-reactive, predictive-reactive, and robust pro-active. In the first strategy, decisions are made locally and jobs are dispatched in real time and according to some dispatching rules or heuristics which prioritize jobs waiting to be processed on a free machine (El-Bouri, 2012, Nurre and Sharkey, 2018, Bektur and Saraç, 2019, Riahi et al., 2019). In the second strategy an initially generated schedule is updated and revised when

Table 1
Summary of the existing literature on APS.

Author/Year	Objective	Mathematical Model	Solution Approach	Products Structure
Yan et al. (2003)	***Minimize:*** Over/under production of finished goods Set-up cost Idle times of work-cells on the line Makespan load deviations among work-cells	MIP	TS	SL
Moon et al. (2004)	***Minimize:*** Total transition time	MIP	GA	SL
Pongcharoen et al. (2004)	***Minimize:*** Earliness costs for all components Earliness costs for all products Tardiness costs for all products	–	GA	ML
Zhang and Yan (2005)	***Minimize:*** Holding costs of the work-in-process inventory Production costs Penalty costs of the over/under produced parts Tardiness time costs Setup costs ***Maximize:*** Earliness time	NMIP	HGA	SL
Chen and Ji (2007b)	***Minimize:*** Cost of production idle time Cost of tardiness penalties Cost of earliness penalties	MIP	–	ML
Chen and Ji (2007a)	***Minimize:*** Cost of production idle time Cost of tardiness penalties Cost of earliness penalties	MIP	GA	ML
Almada-Lobo et al. (2008)	***Minimize:*** Weighted sum of setup times Average inventory levels Number of stock-outs	MIP	VNS	SL
Örnek et al. (2010)	***Minimize:*** Cost of production idle time Cost of tardiness penalties Cost of earliness penalties	MIP	–	ML
Peng et al. (2014)	***Minimize:*** Cost of tardiness penalties Cost of earliness penalties	CP	IB&B	ML
Yan et al. (2015)	***Minimize:*** Production cost Inventory cost Penalty cost of output shortfall Tardiness cost Setup cost	NMIP	AIHGA	SL
An and Yan (2016)	***Minimize:*** Backlogging cost Inventory holding cost Production cost Set-up cost, Earliness cost Tardiness cost Idle-resource cost	MIP	LR	SL
Ozturk and Ornek (2016)	***Minimize:*** Makespan	CP	CPBH	ML
Sukkerd and Wuttipornpun (2016)	***Minimize:*** Tardiness cost Earliness cost Flow time cost	NLP	HGATS	ML
Chen et al. (2017)	***Minimize:*** Over capacity	MIP	TS SA	SL
Aguirre et al. (2017)	***Maximize:*** Total profit of the system	MIP	RH	SL
Chansombat et al. (2019)	***Minimize:*** Total penalty cost	MIP	HDBKH	ML
Moghaddam and Saitou (2019)	***Minimize:*** Cost of production idle time Cost of tardiness penalties Cost of earliness penalties	MIP	GA	ML
Moghaddam and Saitou (2020)	***Minimize:*** Cost of production idle time Cost of tardiness penalties Cost of earliness penalties	MIP	–	ML
Han et al. (2020)	***Minimize:*** Total inventory Production setup cost	FBLDM	PSO	SL
Georgiadis et al. (2021)	***Minimize:*** Inventory cost Backlog cost Changeover cost	MIP	DT	SL
Vieira et al. (2021)	***Minimize:*** Operational cost Production makespan	MIP	DES	SL

MIP: Mixed Integer Programming, ***CP:*** Constraint Programming, ***NMIP:*** Nonlinear Mixed Integer Programming, ***NLP:*** Nonlinear Programming, ***FBLDM:*** Fuzzy Bi-level Decision Making, ***GA:*** Genetic Algorithm, ***HGA:*** Hybrid Genetic Algorithm, ***TS:*** Tabu Search, ***VNS:*** Variable Neighborhood Search, ***IB&B:*** Iterative Branch & Bound, ***AIHGA:*** Alternant Iterative Genetic algorithm, ***LR:*** Lagrangean Relaxation, ***CPBH:*** Constraint Programming Based Heuristics, ***HGATS:*** Hybrid Genetic Algorithm Tabu Search, ***SA:*** Simulated Annealing, ***RH:*** Rolling Horizon, ***HDBKH:*** Hybrid Discrete Bat Krill Herd-based, ***PSO:*** Particle Swarm Optimization, ***DT:*** Decomposition Technique, ***DES***: Discrete Event Simulation, ***SL:*** Single Level, ***ML:*** Multi-Level.

certain disruptions occur in the system (Duenas and Petrovic, 2008, Tighazoui et al., 2021). The last strategy is focused on building predictive schedules which could satisfy certain performance requirements predictably (Vieira et al., 2003).

In most of the previous works in the literature on dynamic scheduling, the predictive–reactive scheduling approach was adopted (Ouelhadj and Petrovic, 2009, Renke et al., 2021). In this regard, the main objective of implemented rescheduling strategies was to minimize some classical shop efficiency measures including make-span (Tang et al., 2016, Li et al., 2017, Wu et al., 2018), total completion time or total weighted completion time (Yang, 2007, Luo et al., 2018), the average or total flow time (Ozlen and Azizoğlu, 2011), and the average tardiness, percentage tardy or total weighted tardiness (Kunnathur et al., 2004, Kaplan and Rabadi, 2015).

When only the above efficiency criteria are considered, new schedules may significantly deviate from the original schedule and therefore, planning activities which were based on the original schedule can be negatively affected. Hence, creating robust predictive–reactive schedules that consider schedule stability measures as well as efficiency measures has become a subject of interest in the dynamic scheduling domain (Leus and Herroelen, 2005, Minguillon and Stricker, 2020).

Implementing rescheduling techniques in the dynamic environment of today's manufacturing systems is clearly inevitable. However, in production environments where different models of a certain product family are manufactured (assembled) in multiple levels, considering the possibility of reassigning (re-pegging) the WIP produced prior to occurrence of a disruptive event as well as rescheduling has been shown to be even more effective (Moghaddam and Saitou, 2019). To the best of our knowledge, Moghaddam and Saitou (2019) and Moghaddam and Saitou (2020) are the only papers that have investigated the effect of joint rescheduling/repegging approach through an exact mathematical formulation. In (Moghaddam and Saitou, 2019) it is assumed that all actions are *reactive* and re-pegging and re-scheduling are done after a new order arrives unexpectedly. In this work, however, a predictive/reactive scheduling approach combined with the re-pegging strategy is adopted to solve the APS problem for products with mixed-model, multi-level structure.

3. Problem description and methodology

In this paper, we consider simultaneous repegging/rescheduling problem under disruptions caused by new orders' arrival. Scheduling periods are defined based on the expected arrival time of each order, *i.e.*, the beginning of each scheduling period coincides with the arrival of an order. During each period, a *predictive/reactive* approach is taken towards scheduling. The proposed approach is considered *predictive* since it involves scheduling for the order at hand and the remaining orders based on their expected arrival time. It is also considered *reactive* as it involves rescheduling and re-pegging of WIP that has already been produced during previous periods. The most important assumptions regarding products, production process, and orders are discussed in the following sub-section.

3.1. Assumptions

The assumptions about the products are as follows:

- Final products have multi-level structures represented by product structure trees (PST) (shown in Fig. 1) Final product (P2) in Fig. 1 for example, has one subassembly (P4), and that subassembly is composed of two parts (P5, P6)
- Each node in a product structure tree can be either a subassembly, (if non-terminal) or a final product (if terminal). In either case the nodes are referred to as *items* in this paper for simplicity

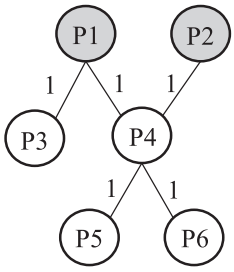


Fig. 1. A simple example of a product structure tree (PST).

- Labels (1′s) on each edge of the PST show how many child items are required for producing one parent item. For example, one P3 and one P4 (child items) are needed to produce one P1 (the parent item)

Furthermore, the following assumptions are made about the production process:

- Production of a parent item cannot be finished unless all its child items are produced. For example, in the PST of Fig. 1, P1 cannot be produced until P3 and P4 are both manufactured
- Operations are non-preemptive and each item can only be processed on one machine
- Each machine can process one item at a time
- Based on the processing time of each item (including setup and transfer time), machines uninterruptedly work for 8 h a day
- In case of an interruption in the production schedule, the production must restart (from the beginning) for those items that are not completely produced

Finally, the orders are placed based on the following assumptions:

- For each order, product model (specified by the item ID in the PST, for example P1), required quantity, due date, and tardiness penalties are given
- Each order consists of only a single part type (for example, only P1 or P2 is ordered)
- The first order is placed at the beginning of the scheduling period and other orders arrive following exponential distributions with different means (the order arrival time of P1 can follow an exponential distribution with mean μ_1 and the order arrival time of P2 can follow an exponential distribution with mean μ_2)
- The number of scheduling periods (The number of orders to be placed) are known in advance
- The production system is empty and idle at the beginning of planning

3.2. Mathematical model

In this section, the proposed MIP model for the predictive-reactive scheduling with dynamic pegging is presented. This model is based on the MIP model for multi-level production scheduling presented by Chen and Ji, 2007b and its extensions with dynamic pegging presented in (Moghaddam and Saitou, 2019; Moghaddam and Saitou, 2020). As in (Moghaddam and Saitou, 2020), the current proposed model assumes exponential arrival time for new orders and predictive scheduling can be conducted prior to each new order arrival. The required parameters and the defined variables for the model are summarized below:

Input Parameters:

T	Total number of scheduling periods
M	Total number of machines
O	Total number of orders
P_i	Item ID of the end product for order i
A_p	Set of child items of item p ($p = 1, \ldots, P_i$)

(*continued on next page*)

(*continued*)

A'_p	Set of parent items of item p ($p = 1, \ldots, P_i$)
$Edge_{pq}$	Number of item p required for one parent item q ($p, q = 1, \ldots, P_i$)
N_{ip}	Number of item p ($p, q = 1, \ldots, P_i$) required in order i ($i = 1, \ldots, O$)
λ_φ	Beginning time of period φ ($\varphi = 1, \ldots, T$) that follows exponential distribution with mean μ_φ
$Q_{i\varphi}$	Quantity of order i ($i = 1, \ldots, O$) in period φ ($\varphi = 1, \ldots, T$)
$DD_{i\varphi}$	Due date (in hours) of order i ($i = 1, \ldots, O$) in period φ ($\varphi = 1, \ldots, T$)
TC_i	Tardiness cost per hour for order i ($i = 1, \ldots, O$)
I	Cost of idle time per hour
t_{ipk}	Processing time required by item p ($p = 1, \ldots, P_i$) of order i ($i = 1, \ldots, O$) on machine k ($k = 1, \ldots, M$)
α	A large positive number

Decision variables:

$S_{ipk\varphi}$	Production start time of item p ($p = 1, \ldots, P_i$) of order i ($i = 1, \ldots, O$) on machine k ($k = 1, \ldots, M$) in period φ ($\varphi = 1, \ldots, T$)
$x_{ipjqk\varphi}$	Binary decision variable. Equals to 1 if item p ($p = 1, \ldots, P_i$) of order i ($i = 1, \ldots, O$) precedes item q ($q = 1, \ldots, P_i$) of order j ($j = 1, \ldots, O$) on machine k ($k = 1, \ldots, M$) in period φ ($\varphi = 1, \ldots, T$), and 0 otherwise
$y_{ip\varphi}$	Available produced quantity of item p ($p = 1, \ldots, P_i$) assigned to order i ($i = 1, \ldots, O$) in period φ ($\varphi = 1, \ldots, T$)
$u_{ip\varphi}$	Binary decision variable. Equals to 1 if arrival time of the order in period φ ($\varphi = 1, \ldots, T$) precedes the start time of part p ($p = 1, \ldots, P_i$) of order i ($i = 1, \ldots, O$) and 0 otherwise
C_φ^{max}	Production make span in period φ ($\varphi = 1, \ldots, T$)
$C_{i\varphi}$	Production completion time of order i ($i = 1, \ldots, O$) in period φ ($\varphi = 1, \ldots, T$)
$L_{i\varphi}$	Number of tardy hours for order i ($i = 1, \ldots, O$) in period φ ($\varphi = 1, \ldots, T$)
$NP_{ip\varphi}$	Quantity of item p ($p = 1, \ldots, P_i$) that should not be produced for order i due to assigning item p's parent to order i ($i = 1, \ldots, O$) in period φ ($\varphi = 1, \ldots, T$)
$APb_{ip\varphi}$	Available produced quantity of item p ($p = 1, \ldots, P_i$) of order i ($i = 1, \ldots, O$) in period φ ($\varphi = 1, \ldots, T$)
$AP_{p\varphi}$	Available produced quantity of item p ($p = 1, \ldots, P_i$) in period φ ($\varphi = 1, \ldots, T$)

The MIP model for all scheduling periods can hence be formulated as shown in (1). In the optimization model, the objective (Eq. (1)) is to minimize sum of the idle time cost of the machines and the tardiness penalty of the orders, during all scheduling periods. Model constraints can be categorized as follows:

3.2.1. Constrains related to scheduling/rescheduling

Constraint (2) indicates that the production make span in each scheduling period must be greater than the production completion time of each order during that period. In Constraint (3) the processing start time of all orders' items on specific machines is set to the arrival time of a new order during a specific period. In other words, the rescheduling start time for all orders on different machines must be greater than or equal to the arrival time of the new order. Constraint (4) is based on the first production assumption and shows that in any scheduling period, processing of an order's item on a machine can only start after the required quantity of its child item(s), from the same order, is processed on any machine. In constraint (5) the completion time of an order is calculated by adding the processing time of the required quantity of an end-product to the start time of that end-product during a certain scheduling period. Constraints (6) and (7) ensure that during a certain scheduling period, only one item is processed on a certain machine at any given time.

$$\min \sum_{\varphi=1}^{T}\left\{ I \times \left\{ M \times C_\varphi^{\max} - \sum_{k=1}^{M}\sum_{i=1}^{O}\sum_{p=1}^{P_i}\left(t_{ipk} \times N_{ip} \times Q_{i\varphi}\right) \right\} + \sum_{i=1}^{O}\left(TC_i \times L_{i\varphi}\right) \right\} \tag{1}$$

Subject to:

$$C_{i\varphi} \leqslant C_\varphi^{\max} \quad \forall i, \varphi \tag{2}$$

$$S_{ipk\varphi} \geqslant \lambda_\varphi \quad \forall i, p, k, \varphi \tag{3}$$

$$S_{ipl\varphi} \geqslant S_{iqk\varphi} + t_{iqk} \times \left(N_{iq} \times Q_{i\varphi} - y_{iq\varphi} - NP_{iq\varphi}\right) \quad \forall q \in A(p), i, p, k, l, \varphi \tag{4}$$

$$S_{iP_ik\varphi} + t_{iP_ik} \times \left(Q_{i\varphi} - y_{iP_i\varphi}\right) = C_i \quad \forall i, k, \varphi \tag{5}$$

$$S_{ipk\varphi} \geqslant S_{jqk\varphi} + t_{jqk} \times \left(N_{jq} \times Q_{j\varphi} - y_{jq\varphi} - NP_{jq\varphi}\right) - \alpha\left(x_{ipjqk\varphi}\right) \quad \forall i, j, p, q, k, \varphi \tag{6}$$

$$x_{ipjqk\varphi} + x_{jqipk\varphi} = 1 \quad \forall i, j, p, q, k, \varphi \tag{7}$$

$$\sum_{i=1}^{O} y_{ip\varphi} = AP_{p\varphi} \quad \forall p, \varphi > 1 \tag{8}$$

$$y_{ip\varphi} \leqslant N_{ip} \times Q_{i\varphi} \quad \forall i, p, \varphi > 1 \tag{9}$$

$$NP_{ip\varphi} = \sum_{q \in A'(p)} y_{iq\varphi} \times Edge_{pq} \quad \forall i, p, \varphi > 1 \tag{10}$$

$$AP_{p\varphi} = \sum_{i=1}^{O} APb_{ip\varphi} - \sum_{i=1}^{O}\sum_{q=1}^{P_i}\left(APb_{iq\varphi} \times EdgeAP_{pq}\right) \quad \forall p, \varphi > 1 \tag{11}$$

$$APb_{ip\varphi} \leqslant N_{ip} \times Q_{i\varphi} \quad \forall i, p, \varphi > 1 \tag{12}$$

$$APb_{ip\varphi} \leqslant y_{ip(\varphi-1)} + NP_{ip(\varphi-1)} + \sum_{k=1}^{M} \frac{\lambda_\varphi - S_{ipk(\varphi-1)}}{t_{ipk}} + \alpha \times \left(1 - u_{ip\varphi}\right) \quad \forall i, p, \varphi > 1 \tag{13}$$

$$APb_{ip\varphi} \leqslant y_{ip(\varphi-1)} + NP_{ip(\varphi-1)} + \alpha \times u_{ip\varphi} \quad \forall i, p, \varphi > 1 \tag{14}$$

$$\lambda_\varphi - S_{ipk(\varphi-1)} \leqslant \alpha \times u_{ip\varphi} \quad \forall i < \varphi, p, k, \varphi > 1 \tag{15}$$

$$\lambda_\varphi - S_{ipk(\varphi-1)} \geqslant -\alpha \times \left(1 - u_{ip\varphi}\right) \quad \forall i < \varphi, p, k, \varphi > 1 \tag{16}$$

$$C_{i\varphi} - DD_{i\varphi} \leqslant L_i \quad \forall i, \varphi \tag{17}$$

$$C_\varphi^{\max} \geqslant 0 \quad \forall \varphi \tag{18}$$

$$S_{ipk\varphi} \geqslant 0 \quad \forall i, p, k, \varphi \tag{19}$$

$$C_{i\varphi} \geqslant 0 \quad \forall i, \varphi \tag{20}$$

$$L_{i\varphi}, y_{ip\varphi} \in \text{integer} \quad \forall i, p, \varphi \tag{21}$$

$$x_{ipjqk\varphi}, u_{ip\varphi} \in \{0, 1\} \quad \forall i, j, p, q, k, \varphi \tag{22}$$

3.2.2. Constraints related to repegging

Constraints (8), (9), and (10) are concerned with repegging. Since all produced items up to the arrival of a new order are already pegged, through these constraints, reassigning the previously pegged items to different orders become possible. Constraint (8) indicates that in any period, repegging a certain item to other orders must be based on the available produced quantity of that item up to that period of time. In other words, one can peg/re-peg what is already produced, not what is expected to be produced in the future. Constraint (9) sets an upper bound for the assigned quantity of an item to a specific order. In fact, the assigned amount cannot be greater than what is actually required of that item. For example, in Fig. 1, if one P1 is ordered and two P4s are manufactured, only one of them can be assigned to P1.

Constraint (10) indicates that by assigning an available produced item to a certain order, all of its child items are assigned to that order as well. For example, in Fig. 1, if one P4 is assigned to P1, then one P5 and one P6 must also be assigned to P1. In this regard, after pegging a parent item to an order, in constraint (10), all its child items are calculated.

When $\varphi > 1$, in constraints (4), (5), and (6), the amounts $y_{iq\varphi}$ and $NP_{iq\varphi}$ subtracted from required quantity of an item ($N_{iq} \times Q_i$) are those items which have been produced during previous scheduling periods and therefore, could have been re-pegged to other orders. In these cases, it is clear that there is no longer a need for a certain item of a specific order to be produced (again).

3.2.3. Constraints related to WIP calculation:

Constraints (11)–(14) are concerned with calculating the exact amount of all produced items up to a certain point in time, i.e., arrival of a new order. In constraint (11), the available produced amount of each item at the beginning of each scheduling period is calculated. As shown in Fig. 2, at the arrival time of a new order (λ_φ) the necessary amount produced of each pegged item can be in one of the following states:

State 1: The total required amount of the pegged item is available (either the pegged item has been produced or another produced item was re-pegged) (item *A* in Fig. 2).

State 2: The total required amount of the pegged item is not available and the new order arrives in the middle of its production (items *B* and *C* in Fig. 2).

State 3: The required amount of the pegged item is not yet produced (item *D* in Fig. 2).

It is clear that at any given moment, each item can only be in one of the above sates (i.e., a new order arrives either before, after, or in the middle of item's production); constraints (15) and (16) ensure the above statement; In the first state, $\lambda_\varphi > S_{ipk(\varphi-1)}$ and therefore, $u_{ip\varphi} = 1$. In this state, if the pegged item is produced as planned, constraint (12) holds. In the second state, $\lambda_\varphi > S_{ipk(\varphi-1)}$ as well, and $u_{ip\varphi} = 1$. Here, the pegged item is in the middle of production and the produced amount at the time of disruption is calculated by constraint (13). In the third state, $\lambda_\varphi < S_{ipk(\varphi-1)}$, $u_{ip\varphi} = 0$ and constraint (14) holds, indicating that item p for the ith order is not yet produced. However, similar items may have been produced in previous periods and re-pegged to order i and therefore argument $y_{ip(\varphi-1)} + NP_{ip(\varphi-1)}$ is added to constraint (14) as well.

Tardiness (in hours) is measured by constraint (17) and finally, the domains of all variables are defined in constraints (18)–(22). *Algorithm* (1) (shown in Table 2) further elaborates on the proposed predictive/reactive approach for solving APS problems when dynamic pegging is implemented. The step by step methodology is explained considering the MILP formulation presented in (1) and the relevant constraints.

4. Results

A simple example is presented in this section to better demonstrate the model's capability. This example is based on the product with the PST shown in Fig. 1. The input values for the model are given in Table 3. There are two different production machines, whose processing times for each item are presented in Table 4.

According to the presented information in Table 3, The first order arrives at $T_1 = 0$. The second and third orders arrive following exponential distributions with means $\mu_2 = 4$ and $\mu_3 = 8$ (hours) respectively. The first and third orders are expected to be for item P1 and the second order is expected to be for item P2. All orders have the quantity of 1. The first order has the due date of 10 and the second and third orders have due dates of 5 h. Clearly, orders' due dates (in hours) are based on each order's expected arrival time; for example, the first order is due at the 10th hour since it is placed at $T_1 = 0$, but the second order is due at the 9th hour as it is expected to arrive at $T_2 = 4$. In other words, the deadline for each order is calculated by adding order's due date to the order's arrival time.

The computing device used in this paper is a personal computer with 2.6 GHz Intel(R) Core(TM) i7 6500U and 16.00 GB RAM, using the Microsoft Windows 10 (64 bit) operation system. The general purpose optimization modeling package GAMS v.24.1.3 with high performance solver CPLEX are used as the modeling tool in the numerical example. GAMS is a high-level modeling system for large scale complex optimization problems. This software embeds CPLEX solver that is capable of finding solutions to linear, mixed integer, and quadratically constrained programming problems. The above numerical example was solved in 11

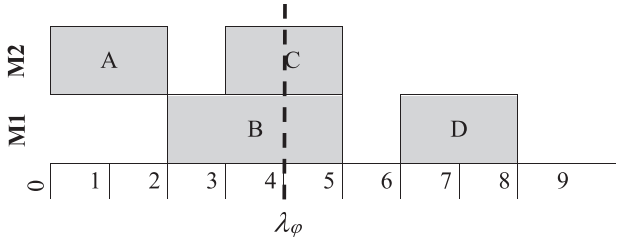


Fig. 2. Arrival of a new order at time λ_φ.

Table 2
The proposed methodology for solving problems through a predictive/reactive approach considering repegging of WIP.

Algorithm (1)

Input: Model data (all inputs of the proposed mathematical model)
Output:A complete production schedule
Begin
  $\varphi \leftarrow 1$; // Set φ to 1
  Solve the model; // Equations (8)–(16) are not considered while solving the model in this step
  A primary schedule is proposed based on the predicted arrival time of all orders
  For $\varphi = 2$ **to** T **do**// When a new order arrives
    Calculate all the produced items before φ; // Equations (11)–(16)
    Simultaneously:
      Re-peg the previously pegged items if necessary; // Equations (8)–(10)
      Reschedule tasks based on predicted arrival time of the remaining orders; // Equations (2)–(7)
  End
End

Table 3
Input values for the simple example.

T	3
M	2
O	3
P_i	$P_1 = P1, P_2 = P2, P_3 = P1$
A_p	$A_1 = \{3, 4\}, A_2 = \{4\}, A_3 = \{\varnothing\}, A_4 = \{5, 6\}, = A_5 = A_6 = \{\varnothing\}$
A'_p	$A_1 = A_2 = \{\varnothing\}, A_3 = \{1\}, A_4 = \{1, 2\}, A_5 = A_6 = \{4\}$
$Edge_{pq}$	$Edge_{31} = Edge_{41} = Edge_{42} = Edge_{54} = Edge_{64} = 1$
N_{ip}	$N_{11} = 1, N_{12} = 0, N_{13} = 1, N_{14} = 1, N_{15} = 1, N_{16} = 1$ $N_{21} = 0, N_{22} = 1, N_{23} = 0, N_{24} = 1, N_{25} = 1, N_{26} = 1$ $N_{22} = 1, N_{32} = 0, N_{33} = 1, N_{34} = 1, N_{35} = 1, N_{36} = 1$
λ_φ	$\lambda_1 = 0, \lambda_2 = 4, \lambda_3 = 8$
$Q_{i\varphi}$	$Q_{11} = Q_{21} = Q_{31} = Q_{21} = Q_{22} = Q_{23} = Q_{31} = Q_{32} = Q_{33} = 1$
$DD_{i\varphi}$	$DD_{11} = DD_{12} = DD_{13} = 10, DD_{21} = DD_{22} = DD_{23} = 5, DD_{31} = DD_{32} = DD_{33} = 5$
TC_i	$TC_1 = TC_2 = TC_3 = 8$
I	6
t_{ipk}	Presented in Table 4
α	100,000,000

Table 4
Machine processing times for the items.

Items	Machine number	Processing Time (hours)
P1	M1	2
P2	M1	2
P3	M2	1
P4	M1	1
P5	M2	1
P6	M2	1

s and the obtained solution to the example is shown in Fig. 3.

The presented tables in Fig. 3 show the optimal solutions in the form of Gantt charts of each scheduling period. In these charts the labels in each cell indicate the order ID, followed by item ID (for example, label "2.6" in the top-left cell of Fig. 3.(a) means that item 6 for the 2nd order is processed on machine M2 during the 1st scheduling period(. As shown in Fig. 3.(a) during the first scheduling period, based on the predicted arrival time of each order, an optimal schedule is presented. In this schedule, all products are produced in 11 h and all orders meet their due dates.

Fig. 3(b) and (c) show the optimal schedules after the 2nd and 3rd orders arrive at $t = 4$ and $t = 8$ respectively. Since the arrival time of these orders (as well as other related information including order type, due date, quantity and tardiness penalties) are exactly as predicted, the proposed schedule during the first scheduling period does not change and neither does the pegging of items to different orders. However, since the second order arrives at $t = 4$, production of item 2 of order 2 is interrupted and therefore the production make span of all orders would become an hour longer in second and third scheduling periods (12 instead of 11 h). Since all orders meet their due dates, the calculated cost during each scheduling period is solely the cost of idle times of the machines.

4.1. Predictive-reactive vs reactive-only scheduling with dynamic pegging

To further investigate the possible advantages of the predictive-reactive (PR) approach, the same example which was presented in the previous section was solved with the reactive-only (RO) approach (fully explained in Moghaddam and Saitou, 2019) as well. Table 5 summarizes the most important differences between the above approaches while solving the same problem. It must be mentioned that when solving the example problem with the RO approach, parameters $DD_{i\varphi}$ and $Q_{i\varphi}$ are as follows: $Q_{11} = Q_{12} = Q_{13} = Q_{22} = Q_{23} = Q_{33} = 1$, $Q_{21} = Q_{31} = Q_{32} = 0$, $DD_{11} = DD_{12} = DD_{13} = 10$, $DD_{22} = DD_{23} = 5$, $DD_{33} = 5$, and $DD_{21} = DD_{31} = DD_{32} = 0$.

Table 5
Differences between PR and RO approaches.

RO Approach	*PR Approach*
• Scheduling is performed only for the first order at $t = 0$. • Second order arrives at $t = 4$. • Re-scheduling and re-pegging are done based on actual arrival times of first and second orders. • Third order arrives at $t = 8$. • Re-scheduling and re-pegging are done based on actual arrival times of all orders	• Scheduling is performed for all orders at $t = 0$, based on the expected arrival times of the second and third orders. • Second order arrives at $t = 4$. • Re-scheduling and re-pegging are done based on actual arrival times of first and second orders and predicted arrival time of the third order. • Third order arrives at $t = 8$. • Re-scheduling and re-pegging are done based on actual arrival times of all orders.

Fig. 4 shows a comparison between the results obtained from solving the problem with the above two approaches. Fig. 4.(a)–(c) is the same solution in Fig. 3 and is presented here again for the sake of easier comparison).

In the PR approach, during all scheduling periods, planning and scheduling are done for all three orders (expected and actual). However, since in the RO approach no knowledge is available of future orders, only the first and the second orders are processed during first and second scheduling periods. Therefore, to make a fair comparison, the average cost of scheduling per period (ACPP = total scheduling cost divided by the number of periods) is also calculated and included in each Gantt chart. For example, in Fig. 4.(a), when PR approach is used, ACCP ($10)

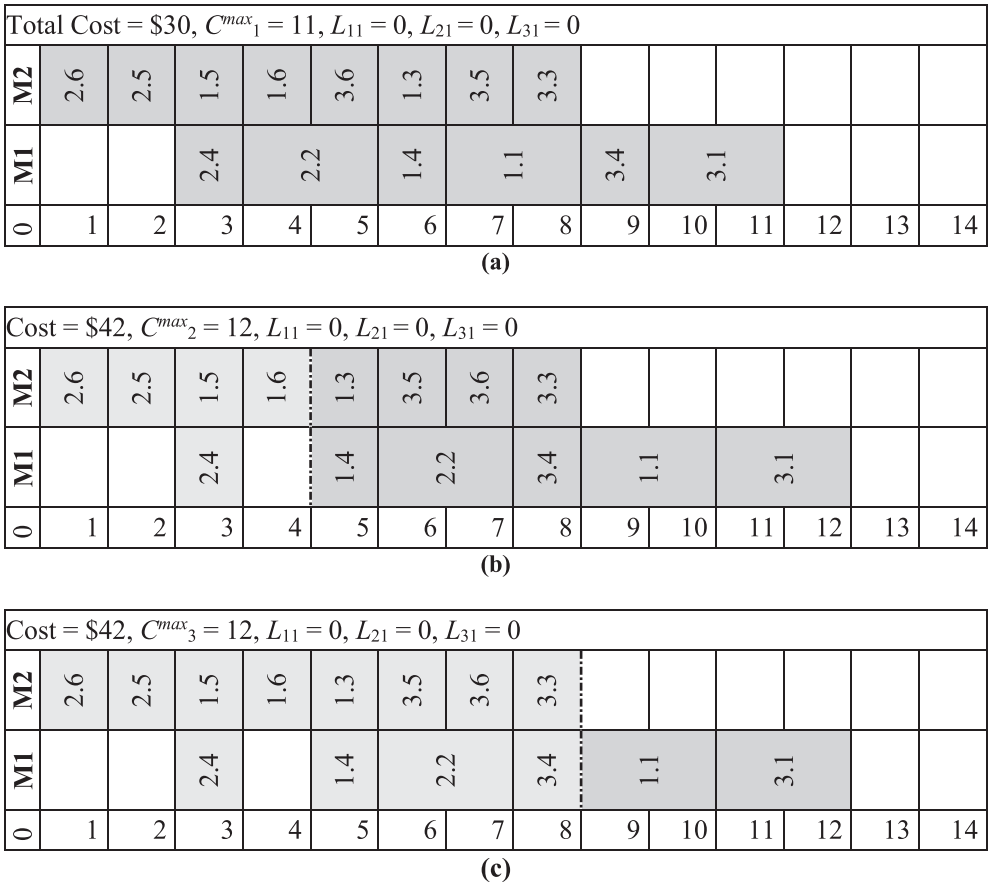


Fig. 3. Optimized schedules for the (a) first ($t = 0$), (b) second ($t = 4$), and (c) third ($t = 8$) scheduling periods.

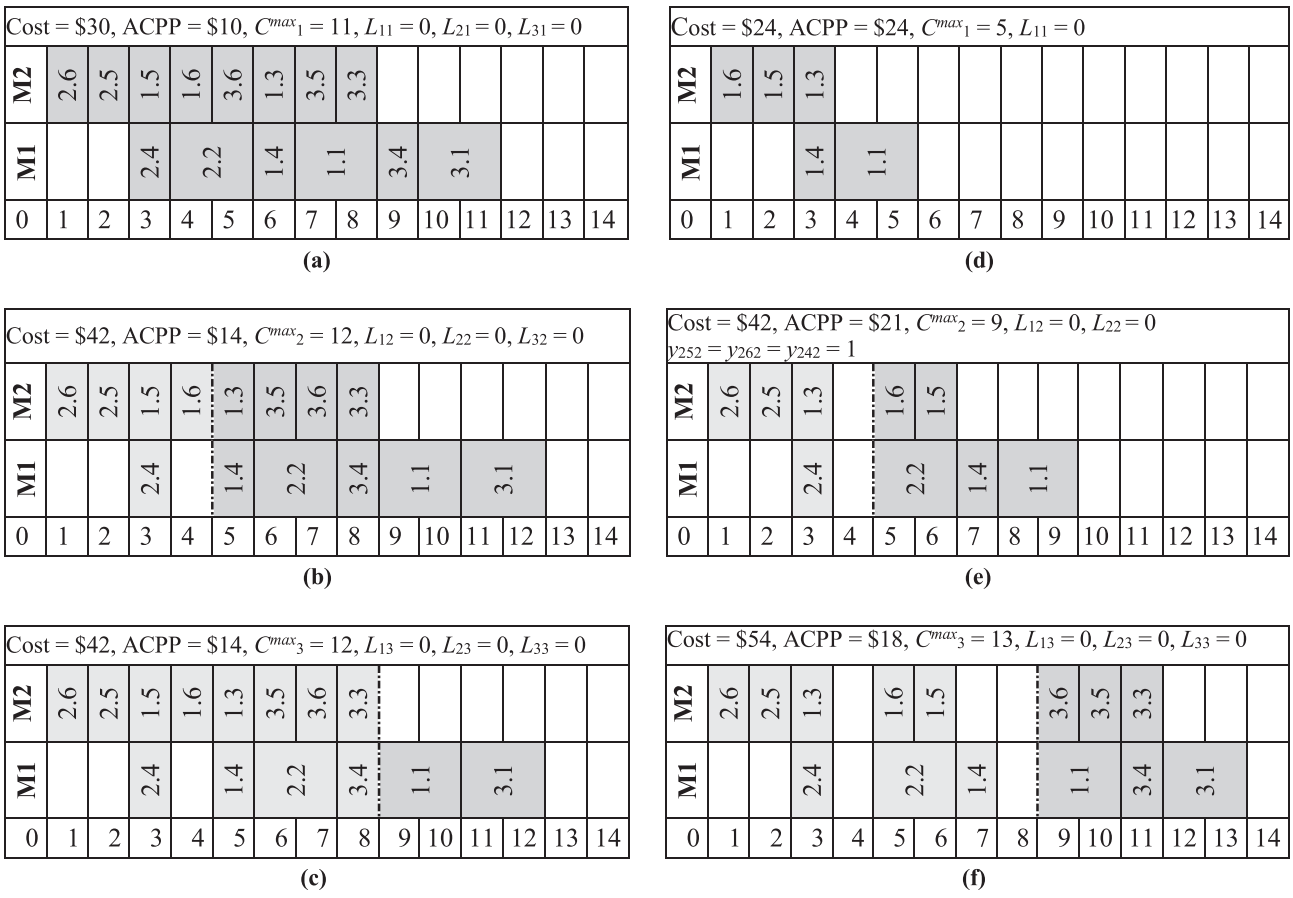


Fig. 4. Optimized schedules with PR approach for the (a) first, (b) second, and (c) third scheduling periods vs the RO approach during the (d) first, (e) second, and (f) third scheduling periods.

is considerably lower than the ACCP calculated when RO approach is taken ($24) (Fig. 4.(d)).

During the second scheduling period (i.e., when the second order arrives at $t = 4$) some repegging is done in the RO approach (since the second order has a tighter due date of 5 h) (Fig. 4.(e)). For example, item 4 (which is already produced) and all its child items are reassigned to the second order. ACPP for PR and RO approaches are $14 and $21 respectively.

During the last scheduling period, (i.e., when the third and last order arrives at $t = 8$), taking the PR approach yields in lower scheduling costs as well (Fig. 4.(c) vs Fig. 4.(f)). In this example, the better performance of PR approach in terms of total cost and average costs per period is as a

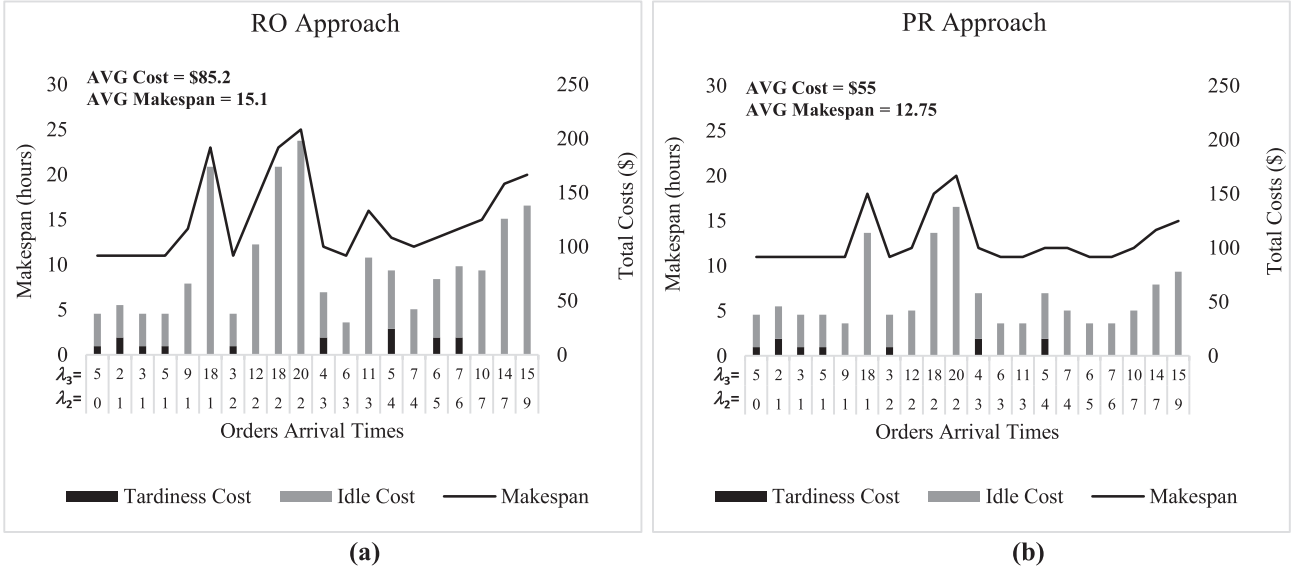


Fig. 5. Changes in cost and makespan when the arrival times of the second and third orders are randomly drawn from exponential distributions with means $\mu_2 = 4$ and $\mu_3 = 8$ respectively using (a) RO and (b) PR approach.

result of the prior information about all orders. This information led to better use of machinery (less idle time of the machines) and even a better chance to meet tight due dates.

In the simple numerical example above, its assumed that arrival time of the second and third orders are exactly as predicted. In the real world however, 100 % accuracy of predictions seldom is the case. To further investigate the effect of adopting PR approach on rescheduling performance, 20 random arrival times were drawn from exponential distributions with means $\mu_2 = 4$ and $\mu_3 = 8$ for the second and third orders respectively (it is assumed that the first order always arrives at $t = 0$). The results are shown in Fig. 5. (a) and Fig. 5. (b) for the RO and the PR approach.

In Fig. 5, the X coordinates, shown by λ_2 and λ_3, are actual random numbers drawn from the above two exponential distributions. In other words, the simple example was solved 20 times with different combinations of actual arrival times assuming the first order always arrives at $t = 0$. As can be seen, for order arrival times that are considerably greater than the expected amount, there exists a difference between the makespan and total cost of the two approaches. A simple *t*-test on the results obtained from the two approaches also indicates that the average total costs differ significantly between the two groups (CI = 95 %, P-Value = 0.033). A more comprehensive analysis of the arrival times' deviation from the expected amount is done in section 4.2.1.

In the next section, to provide better insights on how the two approaches perform in face of inaccurate information, both PR and RO approaches are analyzed and compared while relaxing the constraint on perfect predictions. In other words, sensitivity of each approach to unexpected deviations from predicted arrival times, order quantities, and due-dates is analyzed while addressing different hypothetical scenarios.

4.2. Comparison scenarios

The same numerical example in Section 3 is used here as well. However, in the proposed scenarios, changes are made in some input information such as arrival time (earlier/later arrivals), due-dates (tighter due dates) and order quantity (increase in the order amount or order cancellations).

4.2.1. Scenario 1: Changes in arrival time

As discussed earlier, dealing with stochastic arrival time is a challenging problem. While the PR approach can be more effective when the arrival time predictions are completely accurate, it is important to compare PR and RO approaches and more carefully observe their performance when the predictions are not entirely correct.

4.2.1.1. Sooner than expected arrivals. In this section, it is assumed that both second and third orders arrive earlier than the predicted time. The solution to one of the cases where actual arrival times are changed from 4 to 3 and from 8 to 6 for the second and third orders, is shown in Fig. 6. By comparing the results in Fig. 6.(a)−(c) and (d)−(f) one can see that, both PR and RO approaches produce similar results when arrival times are less than the expected value. In both approaches the final cost (incurred cost in the third scheduling period) is $30, no re-pegging is done, and all orders meet their due dates.

To further analyze the effect of inaccurate data on the performance of PR and RO approaches, all possible scenarios where the arrival times of second and third orders are less than the expected amounts are considered. The results are shown in Fig. 7.(a) and (b) for the RO and the PR approaches.

As can be seen, when both orders arrive sooner than the expected

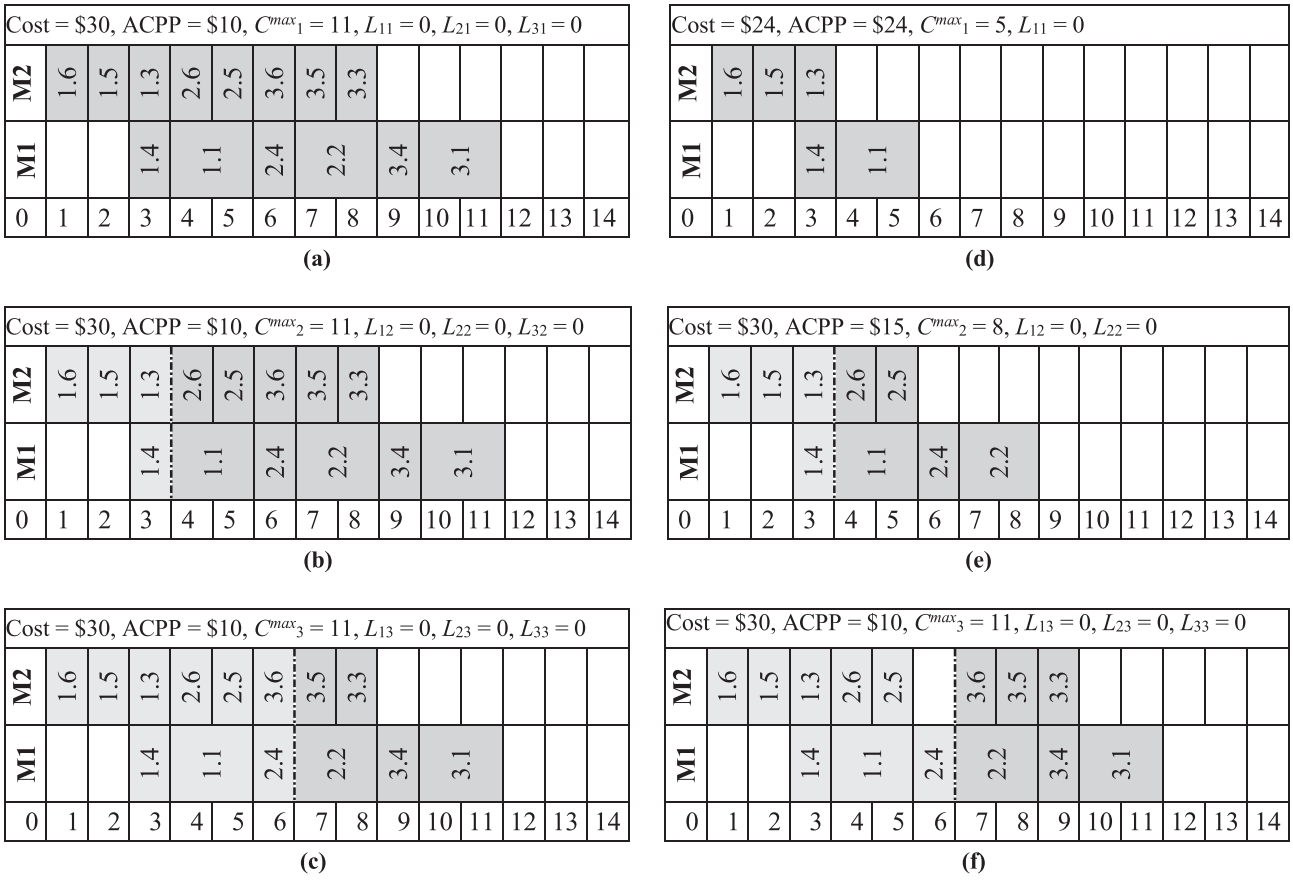


Fig. 6. Optimized schedules with PR approach for the (a) first, (b) second, and (c) third scheduling periods vs the RO approach during the (d) first, (e) second, and (f) third scheduling periods in case of arrivals before the expected time.

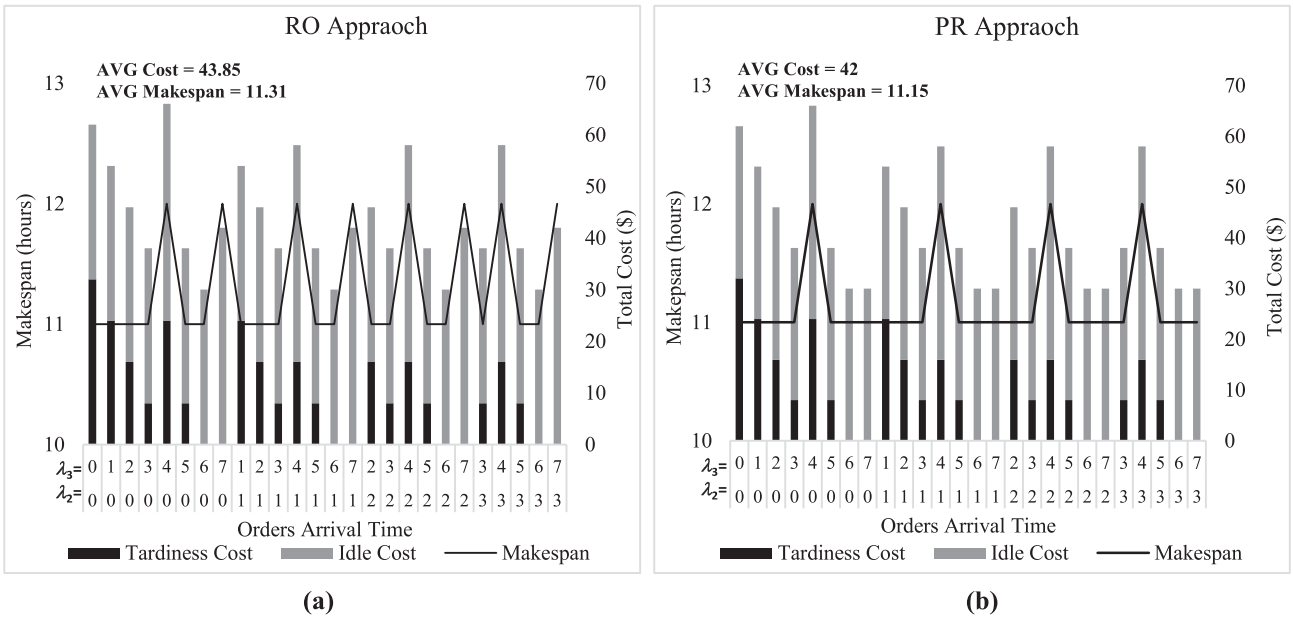


Fig. 7. Changes in cost and makespan when the arrival times of the second and third orders are less than the expected amounts ($\mu_2 = 4$ and $\mu_3 = 8$) using (a) RO and (b) PR approach.

time, there are no significant differences between the average total costs and average make-spans of the two approaches. The *t*-test performed on the results of the two groups also indicates the above statement (CI = 95 %, P-Value = 0.55).

4.2.1.2. Later than expected arrivals. Here, the arrival time of the second and third orders are changed in a way that both orders are placed after the expected time. Fig. 8 shows the solution to one specific case where actual arrival time of the second order is changed from 4 to 6 and actual arrival time of the third order is changed from 8 to 10. As can be seen,

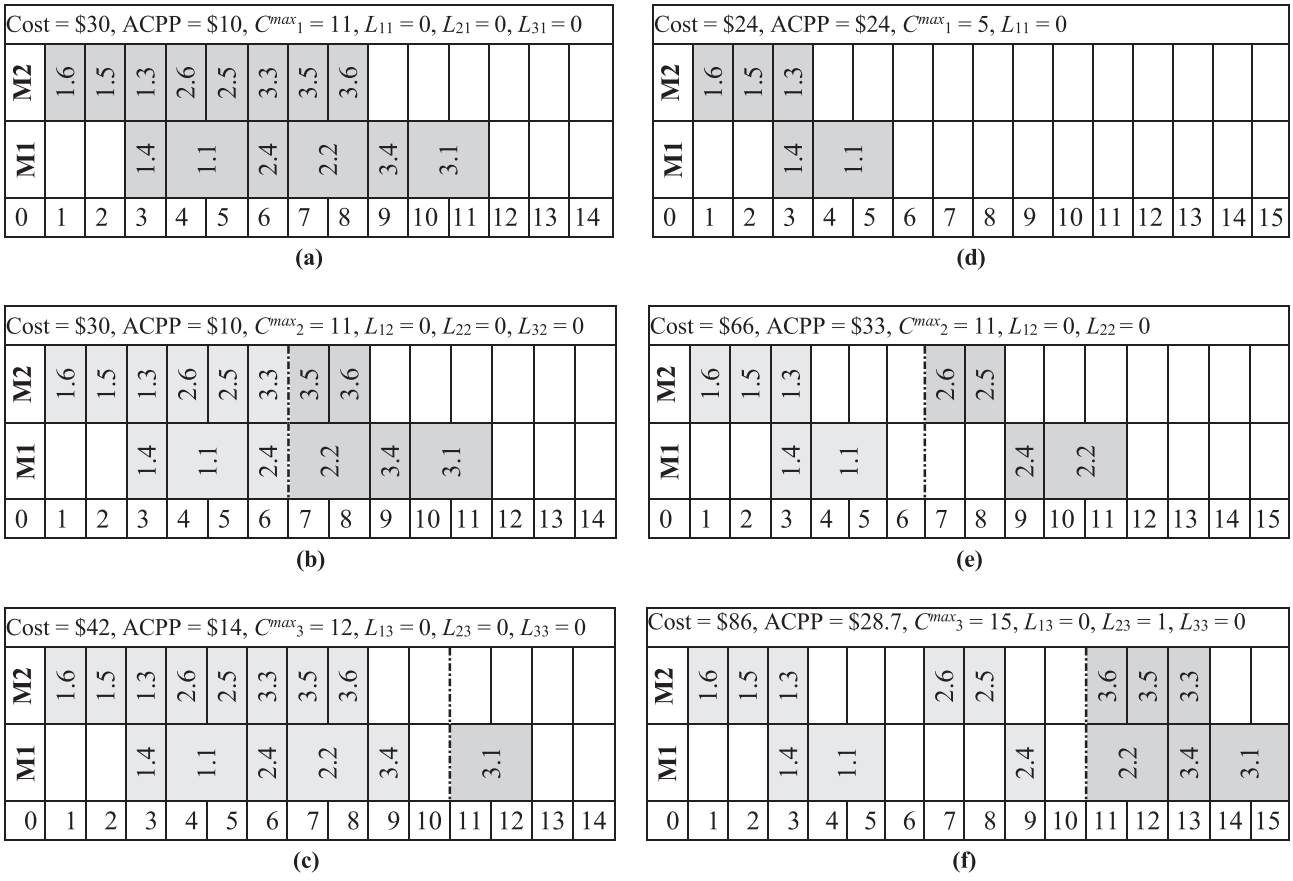


Fig. 8. Optimized schedules with PR approach for the (a) first, (b) second, and (c) third scheduling periods vs the RO approach during the (d) first, (e) second, and (f) third scheduling periods in case of arrivals after the predicted time.

less idle times (compared in Fig. 8.(b) and (e) during the second scheduling period) and shorter make span (compared in Fig. 8.(c) and (f) during the third scheduling period) result in less total cost when the PR approach is taken.

To perform statistical analysis, 20 random numbers are drawn from exponential distributions with means $\mu_2 = 4$ and $\mu_3 = 8$ in such a way that the actual arrival times are more than the expected amount. The results are shown in Fig. 9. (a) and (b) for the RO and the PR approaches. The *t*-test on the obtained results indicates that the average total costs (CI = 95 %, P-Value = 0.013) as well as the average makespan (CI = 95 %, P-Value = 0.018) differ significantly between the two groups of results.

4.2.2. Scenario 2: Changes in Due-Dates

As mentioned before, order arrival time is not the only parameter which can deviate from its expected value. Another important parameter that usually changes after orders are placed is the due date. Larger due dates do not impose serious problems for the production system but when orders' due dates unexpectedly become tighter, the difference between the performance of PR and RO approaches become clear.

Here, the step by step solution to a specific case is presented in Fig. 10. In this case, the due date of the third order changes from 5 to 3 h upon arrival. As can be seen, RO approach basically reacts to the arrival of the third order with a tight due date the same as any other order (since no information about the third order was provided beforehand). However, when the PR approach is taken towards solving the problem, as the third order arrives at $t = 8$, some of its items have already been produced (due to the prior information). Therefore, reacting to the tighter due date costs less than the RO approach ($50 vs $70).

For the second scenario, to perform similar statistical analysis, it is assumed that due dates of all three orders of the proposed numerical example are randomly drawn from uniform distributions in the range of $(0, DD_{i\varphi})$. Therefore, 20 random due dates are generated for each order and the problem is solved with PR and RO approaches for 20 different combinations of due dates. The results are shown in Fig. 11.

The *t*-test on the results obtained from RO and PR approaches indicate a significant difference between the average total costs of the two groups (CI = 95 %, P-Value = 0.003).

4.2.3. Scenario 3: Changes in order quantity

Like the other parameters, increase and decrease in order quantities can affect the performance of PR and RO approaches. In the following, using the same numerical example, the increase in the amount of the third order and then the case of decrease in order quantity in its extreme form (second and third orders cancellation) are investigated.

4.2.3.1. Increase in order quantity. For the third scenario, a single unit increase in the third order's quantity is considered. As shown in Fig. 12.(a)–(c) by expecting a certain amount of P1 in the third order, its production starts before its actual arrival time at $t = 8$. Hence, even though the last order's quantity is increased by one, production of all orders in the final scheduling period is done with minimum costs (including the $36 cost of machines' idle time and $16 penalty on tardy days of 2). However, when there is no information available on the third order, a sudden order of 2 P1s at $t = 8$ can incur more costs to the system.

As can be seen in Fig. 12.(f), even when some previously pegged items are reassigned to the last order, the total cost of rescheduling is still 90 % higher in the RO approach (as shown in Fig. 12.(f), all orders miss their due dates by at least an hour). No statistical analysis is done in this section since increasing the quantity of orders (assuming all other parameters are fixed) is equivalent to decreasing due dates and the scenarios on tighter due dates were previously addressed in Section 4.2.2.

4.2.3.2. Order cancelation. One of the downsides to the PR approach might be the unexpected order cancellation. Sudden order cancellations can lead to unwanted produced items. Even though in products with multi-level structures unwanted produced items can be later re-pegged to previous or new orders, there is a chance that the items produced for the cancelled orders go to waste. In the final scenario, first, it is assumed that the second order (one P2 due in 5 h) is cancelled. In this case, as a result of the similar structure of P1 and P2, the items produced for the second order in the first scheduling period in PR approach (Fig. 13.(a)) are re-pegged to the first and third orders (Fig. 13.(b)).

Fig. 13(f) shows the costs incurred during the last scheduling period using the RO approach. None of the orders are late but the significant difference observed in costs in comparison to the PR approach ($84 vs $48 and $28 vs $16 on average) is the result of idle times on the machines. Since no information about the future orders are available in the

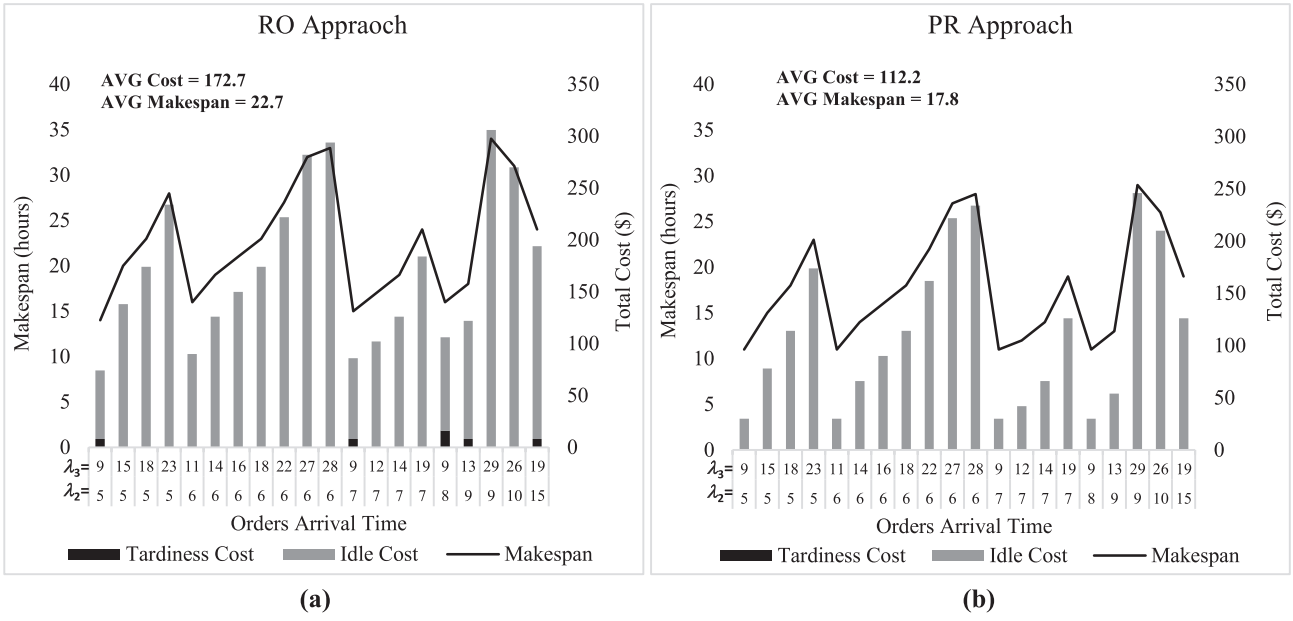


Fig. 9. Changes in cost and makespan when the arrival times of the second and third orders are more than the expected amounts ($\mu_2 = 4$ and $\mu_3 = 8$) using (a) RO and (b) PR approach.

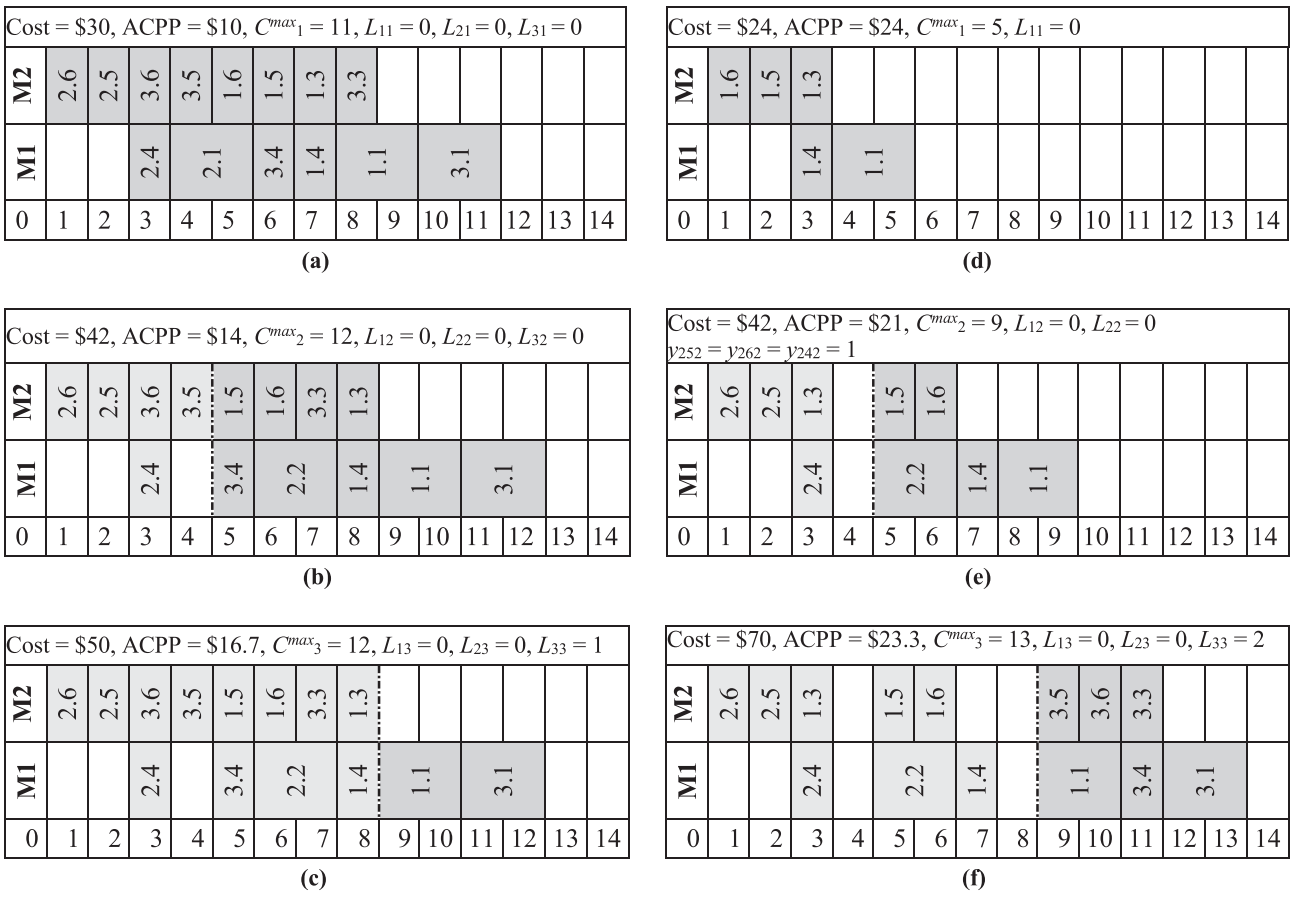


Fig. 10. Optimized schedules with PR approach for the (a) first, (b) second, and (c) third scheduling periods vs the RO approach during the (d) first, (e) second, and (f) third scheduling periods in case of tighter due-dates.

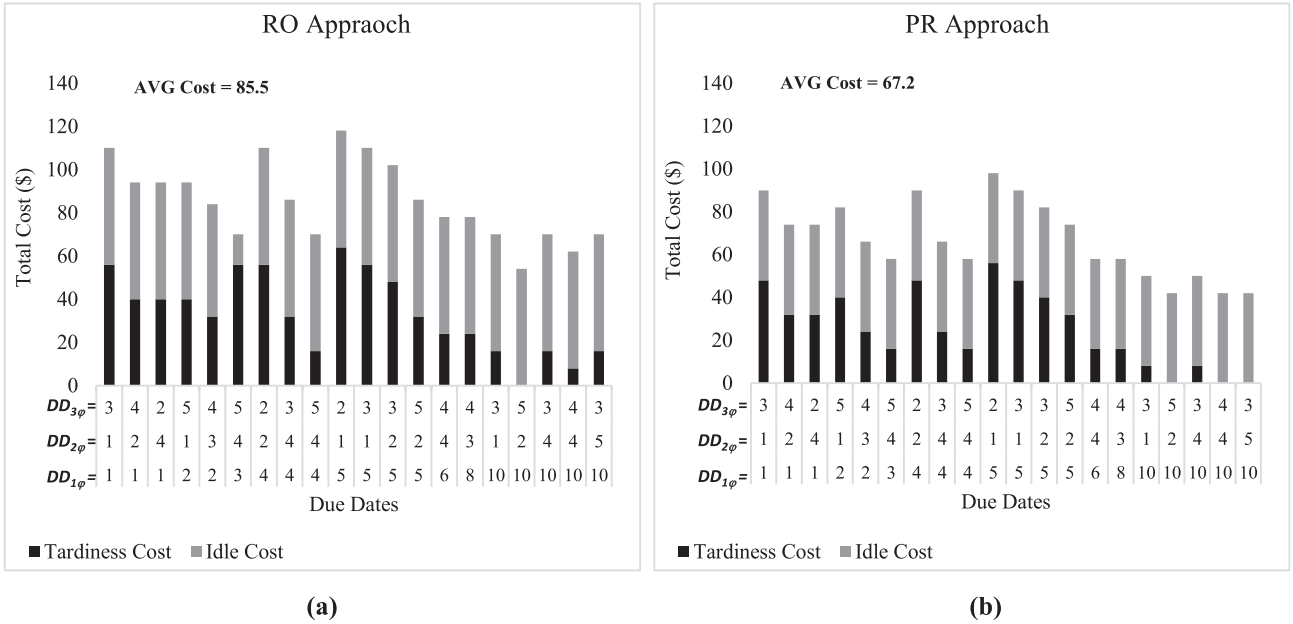


Fig. 11. Changes in total cost when due dates of all orders have been randomly chosen from uniform distributions in the range of (0, $DD_{i\varphi}$) using (a) RO and (b) PR approach.

RO approach, machines remain idle until a new order is placed in the system.

As mentioned earlier, there are certain disadvantages to applying the PR approach when orders can be cancelled. In the example above, if instead of the second order the third order gets cancelled, the results of the two approaches can be quite different (Fig. 14). As can be seen in

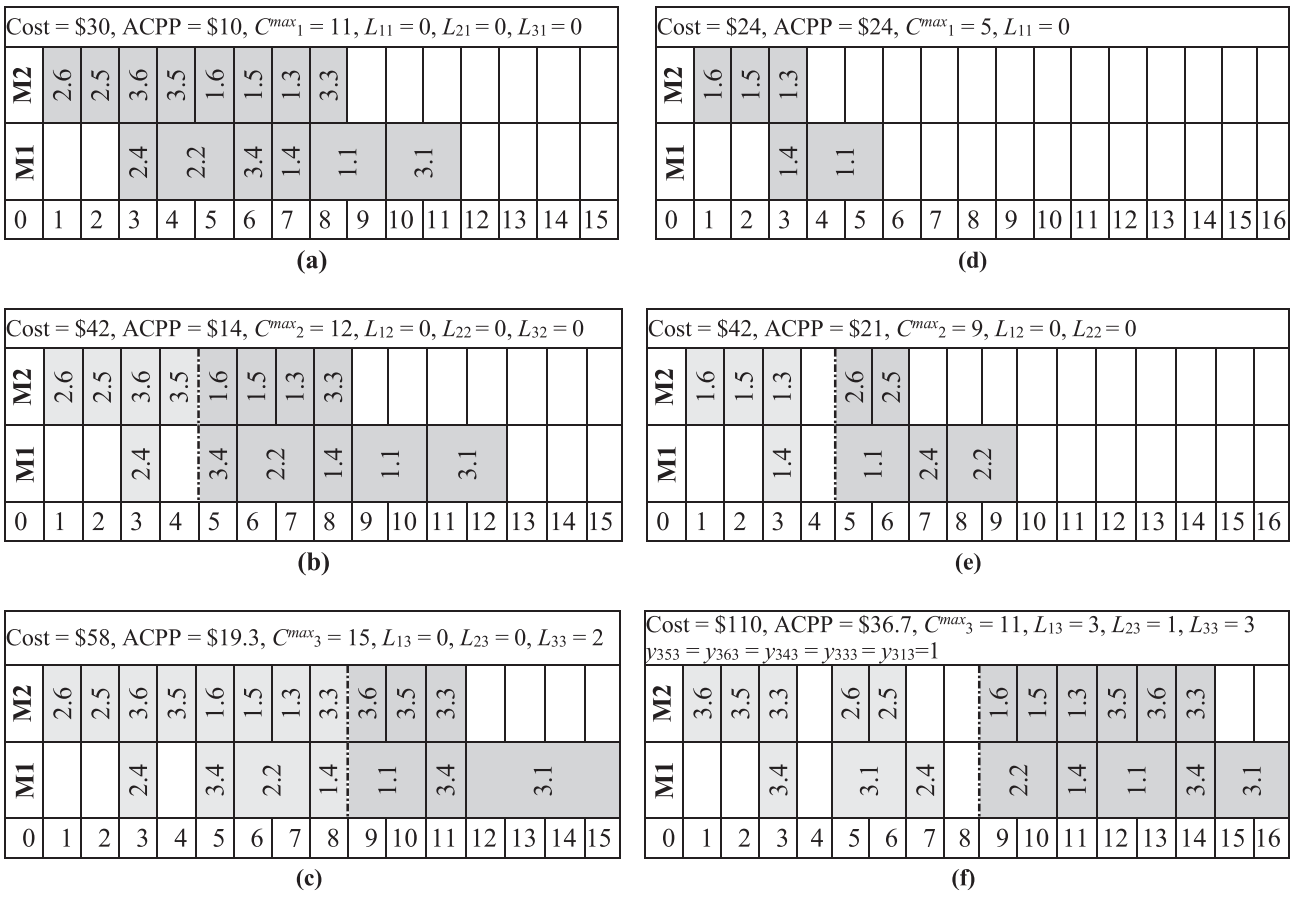


Fig. 12. Optimized schedules with PR approach for the (a) first, (b) second, and (c) third scheduling periods vs the RO approach during the (d) first, (e) second, and (f) third scheduling periods in case of increase in order quantity.

Fig. 14.(c) vs Fig. 14.(f) the total cost of idle time during the last scheduling period is less, in case of adopting the PR approach. However, there are items (colored in black) that are produced in PR approach and cannot be assigned to any other order since the last order of P1 is cancelled.

4.3. Computational time analysis

The APS problem is proved to be NP-hard and the simultaneous rescheduling/re-pegging problem which is addressed in this paper is at least as difficult. The scenario settings in Section 4.2 are all different variations of the simple numerical example in section 4 and the PST shown in Fig. 1. In real world problems, order quantities, the number of scheduling periods (orders), and the number of items in PSTs can be significantly larger. Hence, to gain more insight about the effect of changing (increasing) the most important parameters on problem's complexity, some of them have been increased and the computational time has been further analyzed. The obtained results are shown in Table 6 where larger instances of the simple example are solved with PR and RO approaches. In these examples the due dates for all orders are assumed to be 100 h. Other parameters are exactly the same as what is presented in Table 3.

As shown in Table 6, when there are only 2 orders and hence 2 scheduling periods, the difference between the computational time of RO and PR approaches is not significant and both approaches solve different problems with different order quantities in a short amount of time. Furthermore, the quality of the obtained results in terms of total scheduling cost and makespan is similar in all test cases. When the number of orders is increased by one, the difference between the computational time of RO and PR approaches become significant. 3600 s is the upper bound which was considered in this section due to limitation of the computational resources. In case of 3 orders, RO approach gives exact answers to different instances of the problems with different order quantities in a reasonable amount of time. The results obtained by PR approach are just as good as the ones provided by RO but they are not guaranteed to be optimal which means if the resource capacity was unlimited, the PR approach would probably create less costly solutions to the same problem.

By increasing the number of orders to 4, the MILP formulation in (1) requires significantly more than an hour to create exact results when PR approach is taken. In this case, the time required by RO approach is also considerably increased specifically when order quantity reaches 50. Considering complex product families with more items in the PST will definitely have similar effect on the problem's solving time. As a result of this high complexity level, heuristic and metaheuristic algorithms can be implemented which can result in approximate solutions with acceptable level of quality in a shorter amount of time (Moghaddam and Saitou, 2019).

5. Discussion and implications

In this section, a more comprehensive analysis is provided based on the obtained results in Section 4. The analysis is done with the main intention of providing useful managerial insights for handling realistic larger scale APS problems. It must be emphasized that the following examinations are based on the proposed numerical example of Section 4 and the PST shown in Fig. 1. Also, in the scenario settings of Section 4 only a single parameter is changed for each scenario and all other parameters are assumed to be exactly as predicted. The major implications of our findings can be offered as follows:

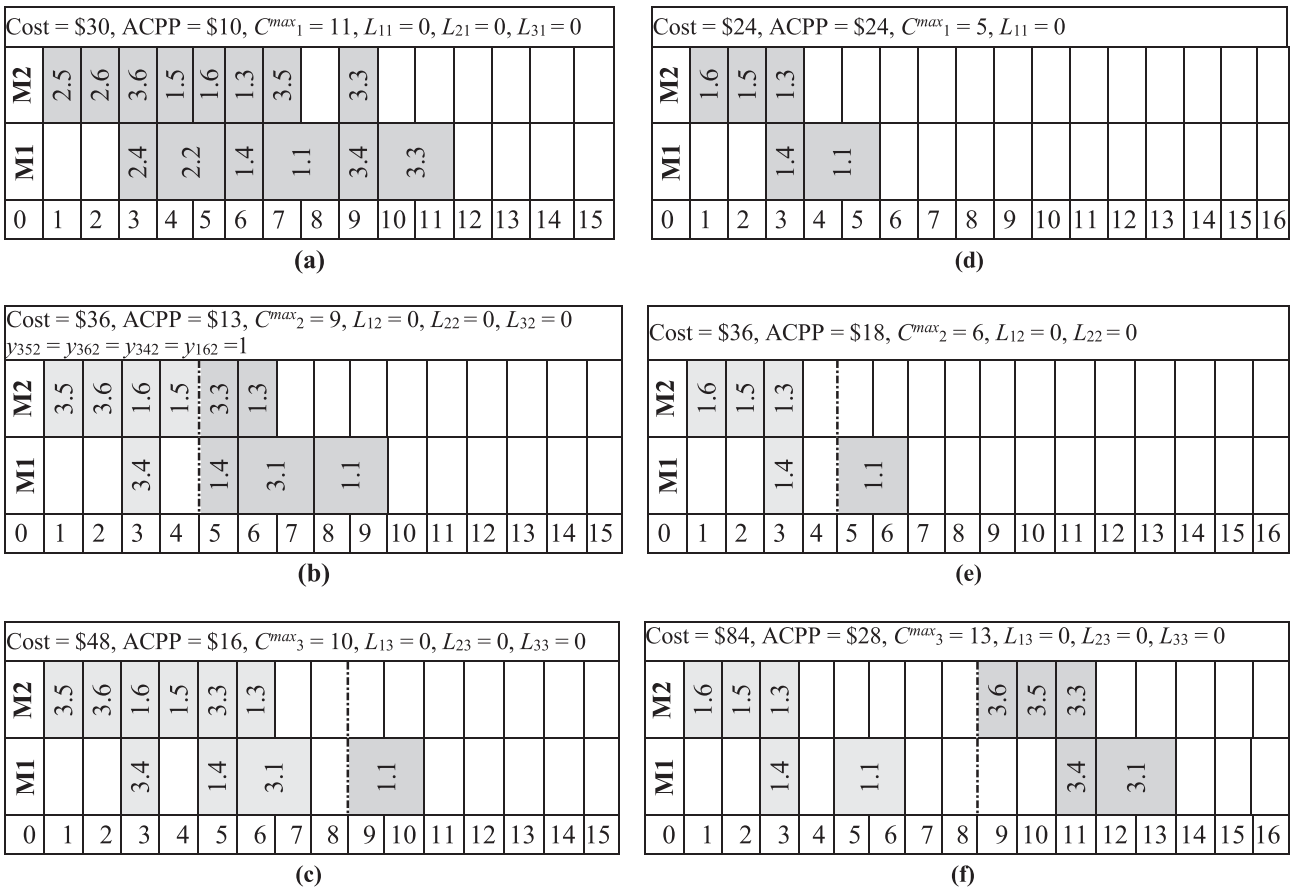


Fig. 13. Optimized schedules with PR approach for the (a) first, (b) second, and (c) third scheduling periods vs the RO approach during the (d) first, (e) second, and (f) third scheduling periods in case of second order cancellation.

- In case of random order arrivals;
  - o If company orders mostly arrive sooner than the expected time, there is no significant advantage in using the PR approach (Fig. 7) and using either approaches is possible.
  - o If company orders are mostly placed after the expected time, there seems to be a significant advantage in using the PR approach due to lower production and scheduling costs and shorter makespan (Fig. 9). However, in terms of total costs, most of the differences observed in the obtained results lie in the idle time cost of the machines (tardiness costs in both approaches are similar). Therefore, if cost of machines' idle time is not quite as important in a manufacturing system (a different cost ratio), the difference between total planning and scheduling costs in the two approaches becomes negligible. Nevertheless, in terms of total makespan PR outperforms the RO approach. The decision on which approach to choose for getting the best results hence depends on the idle time cost of the machines as well as the importance of the shorter makespan.
  - o If company orders are placed in such a random way that it is difficult to establish a pattern in most cases, then taking the PR approach appears to be more cost efficient as well (Fig. 5). Here again, most of the differences in the total scheduling costs are as a result of machines' idle time cost and choosing the better approach depends entirely on the ratio of idle time cost to the tardiness cost. If this ratio is very high, PR approach is better to be implemented and otherwise, selecting either approaches is possible.
- In case of due dates changes where due dates suddenly become tighter than expected, the PR approach performs significantly better in terms of total planning and scheduling costs (Fig. 11). In this case, the differences observed in the obtained results lie in both idle time cost of the machines as well as the tardiness cost of orders. Furthermore, the difference is regardless of the cost of tardiness and cost of idle time per hour; i.e., the number of idle hours and the tardy hours differ significantly (assuming cost of $1 per hour for both tardiness penalties and idle cost the average idle hours plus tardy hours differ significantly between the two groups). Hence, choosing the PR approach results in less costs.
- In case of changes in order quantity,
  - o If orders are usually more than what is expected, the situation becomes similar to when due dates are tight and therefore, adopting the PR approach for production planning and scheduling could be more cost efficient for the production system.
  - o If orders are cancelled very often, taking the RO approach appears to be more logical. In this case, if PR approach is taken, there is always the risk of unwanted WIP and therefore more inventory costs.

Even though the examples proposed in Section 4.2 are kept simple for the sake of better comprehension of the model's output, they provided valuable insights in a reasonable amount of time. The simplicity of these numerical examples was a key factor for being able to compare the proposed two approaches.

While the small statistical hypothesis tests provided in Section 4.2, indicate the significant difference that exist between metrics of interest (total cost and make span), it must be noted that such tests are clearly sensitive to many parameters including sample size and significance level. Furthermore, test results of this sort are strongly dependent to the model inputs and therefore, designed examples and selected parameters can ceratinly affect test results.

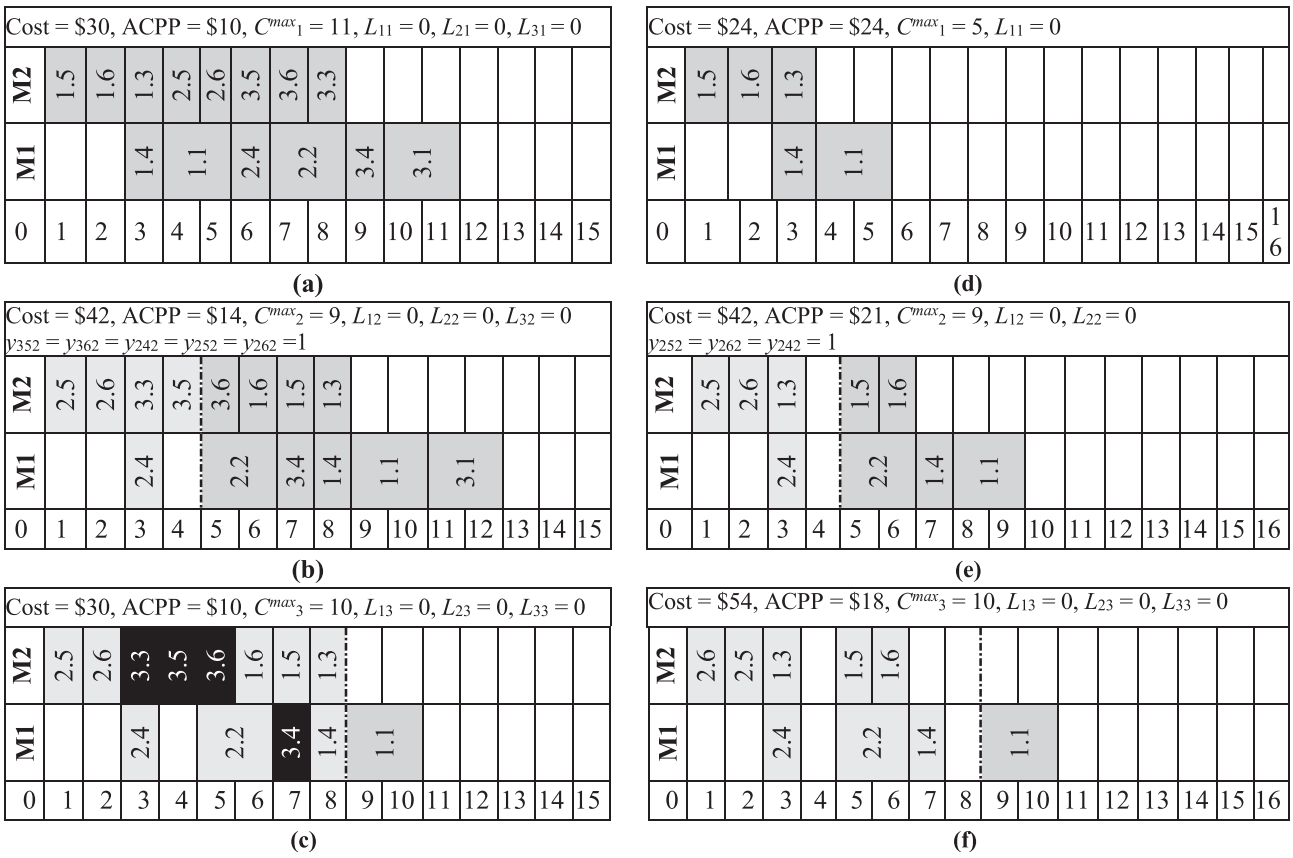


Fig. 14. Optimized schedules with PR approach for the (a) first, (b) second, and (c) third scheduling periods vs the RO approach during the (d) first, (e) second, and (f) third scheduling periods in case of third order cancellation.

Table 6
Computational time analysis of the MILP formulation solved with PR and RO approaches.

# Of Orders	Quantity	RO Approach			PR Approach		
		Total Cost ($)	Makespan (hrs.)	Time (s)	Total Cost ($)	Makespan (hrs.)	Time (s)
2	1	42	9	**0.42**	42	9	**0.52**
	5	150	40	**0.28**	150	40	**0.77**
	10	300	80	**0.17**	300	80	**0.75**
	15	610	120	**0.19**	610	120	**0.48**
	20	1080	160	**0.09**	1080	160	**0.39**
	25	1750	200	**0.17**	1750	200	**0.34**
	50	5100	400	**0.08**	5100	400	**0.3**
3	1	54	13	**0.92**	42	12	**808.41**
	5	150	55	**26.39**	150	55	**3600**
	10	380	110	**14.00**	380	110	**3600**
	15	1130	165	**8.22**	1130	165	**1955**
	20	2040	220	**16.69**	2040	220	**3600**
	25	3150	275	**15.97**	3150	275	**2607**
	50	8700	550	**11.58**	8700	550	**2197**
4	1	120	21	**105.22**	60	16	**3600**
	5	300	80	**16.16**	240	75	**3600**
	10	840	140	**179.08**	840	140	**3600**
	15	2220	210	**590.1**	2220	210	**3600**
	20	3608	280	**196.72**	3600	280	**3600**
	25	5652	358	**4.41**	5300	350	**3600**
	50	13,800	700	**3600**	13,800	700	**3600**

6. Conclusion and future research

This paper presented a new predictive-reactive rescheduling method based on the dynamic pegging concept. Due to the dynamic nature of the manufacturing industry, in most production systems scheduling adjustments are inevitable in response to the unplanned events, such as new order arrival, changes in due dates or order cancellation. In multi-level mixed-model production systems implementing the dynamic pegging approach at the time of rescheduling is quite advantageous since it creates the opportunity of reassigning the already pegged WIP to the existing or new orders. By Incorporating predictive probabilistic parameters into the reactive-only model presented in (Moghaddam and Saitou, 2019) a MIP model is developed for predictive-reactive scheduling of new orders with different product types that arrive following

exponential distribution. A simple example is designed to compare the performance of predictive-reactive rescheduling and the reactive-only rescheduling by changing different input parameters to the MIP model.

The obtained results indicate the better performance of PR over RO in some specific cases including sudden changes in due dates (tighter due dates), random order arrivals (later than expected arrivals) and sudden increase in order quantity. However, when orders arrive sooner than expected, there exists no significant difference between the results obtained by the two approaches, in terms of total scheduling costs and makespan. Also in cases of sudden decrease in amount of orders (order cancellation as the extreme case) adopting the RO approach overall, would probably be less risky as unwanted WIP is not produced.

Many aspects of this work could be further investigated in future research; in a single manufacturing system, it would be valuable to design and analyze more complex examples with various cases of inputs and to show the applicability of the proposed model for a real case in the industry. Integration of the dynamic pegging approach with proactive inventory planning models and more advanced rescheduling techniques is another interesting area of study. More in depth research can be conducted on effects of various kinds of disruptions (resource or job related) in the manufacturing system. Even though some work has been published in this area (Moghaddam and Saitou, 2019), computational time reduction for larger scale real life problems through new heuristics, metaheuristics, or artificial intelligence approaches is yet another important field of study. Analyzing the effect of repegging on robustness of the predictive-reactive scheduling techniques as well as providing a lower bound on the proposed formulation are also interesting subjects of future study.

Examining the influence of dynamic pegging in production planning and scheduling of multi-stage supply chains could yet be another subject of study. Amid coronavirus pandemic, certain attention has been given to flexible, reconfigurable, and resilient supply chain networks which can quickly react to changes and uncertainties. Based on the results of this work, simultaneous pegging and rescheduling could enable fast adaptation to various unpredictable scenarios and the effect of adopting the proposed methodology on different parameters of supply chain planning and scheduling including production costs, storage costs, transportation costs, tardiness costs, or penalty costs for non-delivery can be further investigated.

CRediT authorship contribution statement

Shokraneh K. Moghaddam: Conceptualization, Methodology, Software, Validation, Formal analysis, Writing – original draft. **Kazuhiro Saitou:** Conceptualization, Methodology, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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